serviceDiscovery HTTP/TLS/JSON

Interface Design Description

Service ID: *"serviceDiscovery"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `serviceDiscovery` service of the ServiceRegistry system.

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1 Overview

This document describes the `serviceDiscovery` service interface of the ServiceRegistry system, which provides registration and lookup of service instances: which provider offers which service definition, in which version, through which interfaces and with which security policy.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `serviceDiscovery` service interface

This document provides the Interface Design Description IDD to the *serviceDiscovery – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `serviceDiscovery` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https` (`generic_http` when TLS is disabled), access port 8443 and base path `/serviceregistry/service-discovery`. The `operations` property of the interface lists the operations below with their method and path relative to the base path:

Operation	Method	Path
`register`	POST	`/register`
`lookup`	POST	`/lookup`
`revoke`	DELETE	`/revoke`

2 Service Operations

The interfaces of the `serviceDiscovery` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `serviceDiscovery` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json`. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /serviceregistry/service-discovery/register

Operation: register
Input: ServiceInstanceCreateRequest
Output: ServiceInstanceResponse

Registers a service instance of the requesting system. An existing instance with the same identifier is replaced. Unknown service definitions are created; interfaces are validated against their templates.

A successful call returns `201 Created` with the output in the body.

```
POST /serviceregistry/service-discovery/register HTTP/1.1
Host: serviceregistry:8443
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "serviceDefinitionName": "temperatureReading",
  "version": "1.0.0",
  "expiresAt": "2026-10-22T12:00:00Z",
  "metadata": {
    "unit": "celsius"
  },
  "interfaces": [
    {
      "templateName": "generic_https",
      "protocol": "https",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
          "192.168.0.10"
        ],
        "accessPort": 8080,
        "basePath": "/temperature",
        "operations": {
          "read": {
            "method": "GET",
            "path": "/read"
          }
        }
      }
    }
  ]
}
```

Listing 1: An register invocation.

```
{
  "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
  "provider": {
    "name": "TemperatureSensor",
    "metadata": {
      "location": "hall-1"
    },
    "version": "1.0.0",
    "addresses": [
      {
        "type": "IPV4",
        "address": "192.168.0.10"
      }
    ],
    "device": {
      "name": "Raspberry1",
      "metadata": {
        "location": "hall-1"
      },
      "addresses": [
        {
          "type": "IPV4",
          "address": "192.168.0.10"
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    },
    "createdAt": "2026-10-22T12:00:00Z",
    "updatedAt": "2026-10-22T12:00:00Z"
  },
  "serviceDefinition": {
    "name": "temperatureReading",
    "createdAt": "2026-10-22T12:00:00Z",
    "updatedAt": "2026-10-22T12:00:00Z"
  },
  "version": "1.0.0",
  "expiresAt": "2026-10-22T12:00:00Z",
  "metadata": {
    "unit": "celsius"
  },
  "interfaces": [
    {
      "templateName": "generic_https",
      "protocol": "https",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
          "192.168.0.10"
        ],
        "accessPort": 8080,
        "basePath": "/temperature",
        "operations": {
          "read": {
            "method": "GET",
            "path": "/read"
          }
        }
      }
    }
  ],
  "createdAt": "2026-10-22T12:00:00Z",
  "updatedAt": "2026-10-22T12:00:00Z"
}
```


Listing 2: An register response.

2.2 POST /serviceregistry/service-discovery/lookup?verbose={*verbose*}

Operation: lookup
Input: ServiceInstanceLookupRequest
Output: ServiceInstanceListResponse

Returns the service instances matching all given filters. Under the restricted discovery policy, a requester without direct access only receives instances whose service has the `unrestrictedDiscovery` metadata (the core services). With `verbose=true` the provider and device details are included.

Query parameters: *verbose* (boolean, optional, default `false`). List values are repeated or comma-separated.

A successful call returns `200 OK` with the output in the body.

```
POST /serviceregistry/service-discovery/lookup?verbose=<verbose> HTTP/1.1
Host: serviceregistry:8443
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "instanceIds": [
    "TemperatureSensor|temperatureReading|1.0.0"
  ],
  "providerNames": [
    "TemperatureSensor"
  ],
  "serviceDefinitionNames": [
    "temperatureReading"
  ],
  "versions": [
    "1.0.0"
  ],
  "alivesAt": "2026-10-22T12:00:00Z",
  "metadataRequirementsList": [
    {
      "unit": "celsius"
    }
  ],
  "addressTypes": [
    "IPV4"
  ],
  "interfaceTemplateNames": [
    "string"
  ],
  "interfacePropertyRequirementsList": [
    {
      "unit": "celsius"
    }
  ],
  "policies": [
    "string"
  ]
}
```

Listing 3: An lookup invocation.

```
{
  "entries": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
```

```
        "addresses": [
          {}
        ],
        "device": {
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "count": 1
}
```


Listing 4: An lookup response.

2.3 DELETE /serviceregistry/service-discovery/revoke/*{instanceId}*

Operation: revoke
Input: String

Removes a service instance of the requesting system. Returns 200 if removed, 204 if it did not exist.

Path parameter: *instanceId*.

A successful call returns `200 OK` if the entry was removed, `204 No Content` if there was nothing to remove.

```
DELETE /serviceregistry/service-discovery/revoke/<instanceId> HTTP/1.1
Host: serviceregistry:8443
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 5: An revoke invocation.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.13, which are used to represent things like hashes, identifiers and texts.

3.1 struct Address

A network address with its type.

Object Field	Value Type	Description
"type"	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"address"	String	One network address (host name, IPv4, IPv6 or MAC address).

3.2 struct DeviceResponse

A registered device.

Object Field	Value Type	Description
"name"	String	Device name.
"metadata"	Object	Free-form metadata (JSON object).
"addresses"	Array<Address>	Network addresses with their types.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.4 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

3.5 struct ServiceDefinitionResponse

A service definition.

Object Field	Value Type	Description
"name"	String	Service definition name.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.6 struct ServiceInstanceCreateRequest

Registers a service instance of the requesting system.

Object Field	Value Type	Description
"serviceDefinitionName"	String	Service definition name, camelCase (e.g. `temperatureReading`).
"version"	Version	Version of the service instance (default `1.0.0`).
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Request>	Service interfaces offered by the instance.

3.7 struct ServiceInstanceInterfaceRequest

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface (`NONE`, `CERT_AUTH`, token policies).
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

3.8 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

3.9 struct ServiceInstanceListResponse

List of service instances.

Object Field	Value Type	Description
"entries"	Array<ServiceInstanceResponse>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

3.10 struct ServiceInstanceLookupRequest

Filter for service instances; all fields optional, combined with AND.

Object Field	Value Type	Description
"instanceIds"	Array<String>	Instance identifiers to match.
"providerNames"	Array<String>	Provider system names to match.
"serviceDefinition Names"	Array<String>	Service definition names.
"versions"	Array<Version>	Service versions to match.
"alivesAt"	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
"metadataRequirements List"	Array<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
"addressTypes"	Array<String>	Address types the interfaces must offer (`HOSTNAME`, `IPV4`, `IPV6`, `MAC`).
"interfaceTemplate Names"	Array<String>	Accepted interface templates, e.g. `generic_https`.
"interfaceProperty RequirementsList"	Array<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
"policies"	Array<String>	Interface security policies to match, e.g. `CERT_AUTH`.

3.11 struct ServiceInstanceResponse

A registered service instance.

Object Field	Value Type	Description
"instanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
"provider"	SystemResponse	The provider system.
"serviceDefinition"	ServiceDefinitionResponse	The service definition.
"version"	Version	Version of the service instance.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.12 struct SystemResponse

A registered system.

Object Field	Value Type	Description
"name"	String	System name.
"metadata"	Object	Free-form metadata (JSON object).
"version"	Version	Version of the system.
"addresses"	Array<Address>	Network addresses with their types.
"device"	DeviceResponse	The device hosting the system, if registered.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

3.13 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.
Version	JSON String, semantic version `major.minor.patch`.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “serviceDiscovery – Service Description (ServiceRegistry),” Eclipse Arrowhead, SD 5.2.2, 2026, serviceRegistry-v5.2.2-serviceDiscovery-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	